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CAMERA MONITOR SYSTEM WITH HEADS UP DISPLAY FEATURES

Abstract

An example camera monitor system (CMS) includes a plurality of CMS cameras configured to record images of an environment surrounding a commercial vehicle, a plurality of CMS displays, a heads-up display (HUD), and processing circuitry. The processing circuitry is configured to display image feeds from the plurality of CMS cameras on the plurality of CMS displays, detect an object outside of the commercial vehicle that is moving with respect to the commercial vehicle and with respect to the environment surrounding the commercial vehicle, and display a notification of the object on the HUD. The notification indicates at least one of a direction the object is moving with respect to the commercial vehicle, and a particular one of the plurality of CMS displays the object is predicted to appear on next. A method for a camera monitor system (CMS) is also disclosed.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to a camera monitor system (CMS), and more particularly to a CMS that includes heads up display features.

BACKGROUND

[0002] Vehicle camera systems for mirror replacement or for supplementing mirror views are utilized in commercial vehicles to enhance the ability of a vehicle operator to see a surrounding environment of the commercial vehicle. Camera monitor systems (CMS) utilize one or more cameras to provide an enhanced field of view to a vehicle operator. In some examples, the mirror replacement systems cover a larger field of view than a conventional mirror, or include views that are not fully obtainable via a conventional mirror.

[0003] As the number of displays in a vehicle cabin, and correspondingly the amount of data available to a driver, increases, it may become challenging for a driver to determine where they should focus their attention in a CMS to obtain the most relevant information while maximizing their attention to their primary task of driving.

SUMMARY

[0004] A camera monitor system (CMS) according to an example embodiment of the present disclosure includes a plurality of CMS cameras configured to record images of an environment surrounding a commercial vehicle, a plurality of CMS displays, a heads-up display (HUD), and processing circuitry. The processing circuitry is configured to display image feeds from the plurality of CMS cameras on the plurality of CMS displays; detect an object outside of the commercial vehicle that is moving with respect to the commercial vehicle and with respect to the environment surrounding the commercial vehicle; and display a notification of the object on the HUD. The notification indicates at least one of a direction the object is moving with respect to the commercial vehicle, and a particular one of the plurality of CMS displays the object is predicted to appear on next.

[0005] In a further embodiment of the foregoing embodiment, the processing circuitry is configured to detect the object based on one or more of the image feeds.

[0006] In a further embodiment of any of the foregoing embodiments, the CMS includes at least one object detection sensor that includes at least one of a radar sensor, a lidar sensor, and an ultrasonic sensor, and the processing circuitry is configured to detect the object based on output from the at least one object detection sensor.

[0007] In a further embodiment of any of the foregoing embodiments, the processing circuitry is configured to display a first notification on the HUD based on the object being a VRU, and display a second notification on the HUD based on the object being a non-VRU. The second notification is different than the first notification.

[0008] In a further embodiment of any of the foregoing embodiments, the object is a vulnerable road user (VRU).

[0009] In a further embodiment of any of the foregoing embodiments, the object is a forklift.

[0010] In a further embodiment of any of the foregoing embodiments, the commercial vehicle is a mining vehicle or a construction vehicle, and the object is a vehicle that is not a mining vehicle or a construction vehicle.

[0011] In a further embodiment of any of the foregoing embodiments, the processing circuitry is configured to determine that the object is outside of, but is moving towards, a field of view of a particular one of the plurality of CMS cameras. The notification indicates the particular one of the plurality of CMS displays the object is predicted to appear on next. The particular one of the plurality of CMS displays is configured to display the image feed from the particular one of the

CMS cameras.

[0012] In a further embodiment of any of the foregoing embodiments, the notification includes an indication of how close the detected object is to the commercial vehicle, or an indication of how likely a collision is between the commercial vehicle and the detected object.

[0013] In a further embodiment of any of the foregoing embodiments, the notification includes a plurality of display elements that are activated in a sequence to indicate the direction, or includes a shape that indicates the direction.

[0014] A method for a camera monitor system (CMS) according to an example embodiment of the present disclosure includes utilizing a plurality of CMS cameras to record images of an environment surrounding a commercial vehicle, displaying image feeds from the plurality of CMS cameras on a plurality of CMS displays, detecting an object outside of the commercial vehicle that is moving with respect to the commercial vehicle and with respect to the environment surrounding the commercial vehicle, and displaying a notification of the object on a heads up display (HUD). The notification indicates at least one of a direction the object is moving with respect to the commercial vehicle, and which of the plurality of CMS displays the object is predicted to appear on next.

[0015] In a further embodiment of the foregoing embodiment, the detecting includes detecting the object based on one or more of the image feeds.

[0016] In a further embodiment of any of the foregoing embodiments, the detecting includes detecting the object based on output from at least one object detection sensor. The object detection sensor includes at least one of a radar sensor, a lidar sensor, and an ultrasonic sensor.

[0017] In a further embodiment of any of the foregoing embodiments, the displaying includes displaying a first notification on the HUD based on the object being a VRU, and displaying a second notification on the HUD based on the object being a non-VRU. The second notification is different than the first notification.

[0018] In a further embodiment of any of the foregoing embodiments, the object is a vulnerable road user (VRU).

[0019] In a further embodiment of any of the foregoing embodiments, the object is a forklift.

[0020] In a further embodiment of any of the foregoing embodiments, the commercial vehicle is a mining vehicle or a construction vehicle and the object is a vehicle that is not a mining vehicle or a construction vehicle.

[0021] In a further embodiment of any of the foregoing embodiments, the method includes determining that the object is outside of, but is moving towards, a field of view of a particular one of the plurality of CMS cameras. The notification indicates the particular one of the plurality of CMS displays the object is predicted to appear on next. The particular one of the plurality of CMS displays is configured to display the image feed from the particular one of the CMS cameras.

[0022] In a further embodiment of any of the foregoing embodiments, the notification includes an indication of how close the detected object is to the commercial vehicle, or an indication of how likely a collision is between the commercial vehicle and the detected object.

[0023] In a further embodiment of any of the foregoing embodiments, the indication of what direction the object is moving includes a plurality of display elements that are activated in a sequence to indicate the direction, or includes a shape that indicates the direction.

[0024] The embodiments, examples, and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The disclosure can be further understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:

[0026] FIG. **1** is a schematic front view of a commercial truck with a camera mirror system (CMS) used to provide at least Class II and Class IV views.

[0027] FIG. **2** is a schematic birds-eye view of the commercial truck of FIG. **1** with a CMS providing Class II, Class IV, Class V and Class VI views.

[0028] FIG. **3** is a schematic view of example object detection sensors having associated example object detection areas.

[0029] FIG. **4** is a schematic top view of an example vehicle cabin.

[0030] FIG. **5** is a perspective view of the vehicle cabin of FIG. **4**.

[0031] FIG. **6A** illustrates an example of a detected object moving with respect to a commercial vehicle.

[0032] FIG. **6B** illustrates an example heads up display (HUD) image corresponding to the detected movement of FIG. **6A**.

[0033] FIG. **6C** illustrates another example heads up display (HUD) image corresponding to the detected movement of FIG. **6A**.

[0034] FIG. **7A** illustrates an example of a detected object moving with respect to a commercial vehicle.

[0035] FIG. **7B** illustrates an example heads up display (HUD) image corresponding to the detected movement of FIG. **6A**.

[0036] FIG. **7C** illustrates another example heads up display (HUD) image corresponding to the detected movement of FIG. **6A**.

[0037] FIG. **8** schematically illustrates movement of a commercial vehicle with respect to a detected object.

[0038] FIG. **9** is a flowchart of an example method for a CMS.

DETAILED DESCRIPTION

[0039] Schematic views of a commercial vehicle **10** are illustrated in FIGS. **1-5**. The commercial vehicle **10** includes a vehicle cab or “tractor” **12** for pulling a trailer **14**, where the trailer **14** pivots with respect to the tractor **12** during turns. Although the commercial vehicle **10** is depicted as a commercial truck with a single trailer in this disclosure, it is understood that other commercial vehicle configurations may be used (e.g., different types or quantities of trailers).

[0040] Referring now to FIGS. **1-5**, a pair of camera arms **16A-B** include a respective base that is secured to, for example, the cab **12**. A pivoting arm is supported by the base and may articulate relative thereto. At least one rearward facing camera **20A-B** (generally, camera **20**) is arranged respectively within the camera arms **16A-B**. The exterior cameras **20A-B** respectively provide an exterior field of view FOV.sub.EX1, FOV.sub.EX2 that each include at least one of Class II and Class IV views (FIG. **2**), which are legal prescribed views in the commercial trucking industry.

[0041] The Class II view on a given side of the commercial vehicle **10** is a subset of the class IV view of the same side of the commercial vehicle **10**. Multiple cameras also may be used in each camera arm **16A-B** to provide these views, if desired. Class II (narrow) and Class IV (wide angle) views are defined in European R46 legislation, for example, and the United States and other countries have similar drive visibility requirements for commercial trucks. Any reference to a “Class” view is not intended to be limiting, but is intended as exemplary for the type of view provided to a display by a particular camera.

[0042] Each camera arm **16A-16B** may also provide a housing that encloses electronics, e.g., a controller, that are configured to provide various features of the CMS **15**, and/or one or more object detection sensors (e.g., ultrasonic sensor, radar sensor, or lidar sensor). The camera arms **16A-B** may be mounted either at a roof-mount location over the cab door (as shown), or on a door-

mounted bracket or station, for example.

[0043] If video of Class V and Class VI views is also desired, a camera housing **16C** and camera **20C** may be arranged at or near the front of the commercial vehicle **10** to provide those views (FIG. 2). Camera **20C** has a field of view FOV.sub.EX3.

[0044] A backup camera **20D** may be provided which provides a field of view FOV.sub.EX4. The backup camera **20D** may be mounted at a top/centerline of the trailer **14**, at a bumper/bed level of the trailer **14**, or at a top-corner of the back of the trailer **14**, for example. Alternatively, or in addition to the rear trailer camera, a “fifth wheel camera” **20E** may be provided that is mounted to a rear of the tractor **12** and that provides a field of view FOV.sub.EX5. The fifth wheel camera **20E** may be mounted anywhere between the lateral plane of the fifth wheel fixture and the top/roof edge of the tractor **12**, for example.

[0045] A plurality of CMS displays **18A-D** (which may be LCD displays, for example) are provided for displaying image feeds from the plurality of CMS cameras **20A-E**. In one example configuration, CMS display **18A** displays an image feed from camera **20A**, CMS display **18B** displays an image feed from camera **20B**, and CMS displays **18C** and/or **18D** provide an image feed from cameras **20C** and/or **20D**. A heads-up display (HUD) **18E** is also provided.

[0046] FIG. 3 is a schematic view of example object detection sensors **30A-D**, having associated example object detection areas **32A-D**. The object detection sensors **32** may include ultrasonic, radar, or lidar sensors, for example. As shown in FIG. 3, object detection sensors **32A-B**, which are separate from the cameras **20A-B**, may be provided in the camera arms **16A-B**.

[0047] FIG. 4 is a schematic top view of an example vehicle cabin interior **24**, and FIG. 5 is a perspective view of the vehicle cabin interior **24**. Referring now to FIGS. 4-5 with continued reference to FIGS. 1-3, CMS displays **18A-E**, HUD **18E**, cameras **20A-E**, and object detection sensors **30A-D** are shown. A camera monitor system (CMS) (schematically shown as **15**) includes the camera arms **16A-B** mounted to the outside of the vehicle cab **12** (FIG. 1), the cameras **20A-E**, object detection sensors **30A-D**, CMS displays **18A-D**, and HUD **18E**.

[0048] The CMS **15** includes a CMS electronic control unit (ECU) **22** that includes processing circuitry that supports operation of the CMS **15** and is operatively connected to memory (which may include any one or combination of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, VRAM, etc.)) and/or nonvolatile memory elements (e.g., ROM, hard drive, tape, CD-ROM, etc.). The processing circuitry may include one or more microprocessors, microcontrollers, application specific integrated circuits (ASICs), or the like.

[0049] The CMS ECU **22** is configured to display image feeds from the plurality of CMS cameras **20A-E** on the plurality of CMS displays **18A-D**. The electronic displays **18A-B** are arranged on each of the driver and passenger sides within the vehicle cab **12** on or near the A-pillars **19A-B** to display Class II and Class IV views on its respective side of the commercial vehicle **10**, which provide rear facing side views along the commercial vehicle **10** that are captured by the exterior cameras **20A-B**.

[0050] As discussed above, if video of Class V and Class VI views are also desired, the camera housing **16C** and camera **20C** may be arranged at or near the front of the commercial vehicle **10** to provide those views (FIG. 2). Display **18C** is arranged within the cab **12** near the top center of the windshield may be used to display the Class V and/or Class VI views, which are toward the front of the commercial vehicle **10**, or a backup camera view (from camera **20D** or **20E**) to the driver. Display **18D** is provided in a center console area of the cabin interior **24**, and may be used for other purposes, such as navigation, infotainment, etc. HUD **18E** is used to provide various notifications about detected objects, as discussed in greater detail below.

[0051] If video of Class VIII views is desired, camera housings can be disposed at the sides and rear of the commercial vehicle **10** to provide fields of view including some or all of the Class VIII zones of the commercial vehicle **10**. In such examples, one of the displays **18C-E** may include one or more frames displaying the Class VIII views. The displays **18A**, **18B**, **18C** face a driver region

within the cabin interior **24** where an operator is seated on a driver seat.

[0052] If desired, the camera arms **16A-B** may include conventional mirrors integrated with them as well, although the CMS **15** may be used to entirely replace mirrors. In additional examples, each side can include multiple camera arms, each arm housing one or more cameras and/or mirrors.

[0053] The CMS ECU **22** includes object detection features, and is configured to apply conventional object detection techniques to detect objects outside of the commercial vehicle **10** that are moving with respect to the commercial vehicle **10** and with respect to the environment surrounding the commercial vehicle **10**. The detected objects that meets this criteria (moving with respect to both the commercial vehicle **10** and its surrounding environment) could include another vehicle (e.g., a car, a truck, motorcycle, forklift, construction vehicle (e.g., bulldozer or backhoe), or mining vehicle (e.g., a mine cart)), or a vulnerable road user (VRU) such as a pedestrian, cyclist, or animal, for example. A stationary object, such as a road sign or fire hydrant, although potentially moving with respect to the commercial vehicle **10** (if the commercial vehicle **10** is not stationary), would not be moving with respect to the environment surrounding the commercial vehicle **10**). The CMS ECU **22** is operable to detect objects based on one or more of the image feeds from the CMS cameras **16A-D** and/or from the object detection sensors **30A-D**.

[0054] The CMS ECU **22** is configured to perform the object detection based on one of the image feeds from one of the CMS cameras **20** and/or based on data output from one or more of the object detection sensors **30A-D**.

[0055] The CMS ECU **22** is configured to display a notification of a detected object that meets the object detection criteria on the HUD **18E**. The notification indicates at least one of a direction the object is moving with respect to the commercial vehicle **10**, and which of the plurality of CMS displays **18A-D** the object is predicted to appear on next.

[0056] FIG. **6A** schematically depicts movement of an object **52** with respect to the commercial vehicle **10**, and FIGS. **6B-C** depict an example HUD displays **70A-B** that includes a notification of the detected object and its movement. The object may be a vulnerable road user, for example (e.g., pedestrian, cyclist, etc.).

[0057] FIG. **6A** provides a schematic birds-eye view of the commercial vehicle **10**, and illustrates how a detected object **52** moves with respect to the commercial vehicle **10** from a first position **54A** in front of the commercial vehicle **10** (in a Class VI area) to a second position **54B** on a right side of the commercial vehicle (in a Class V area).

[0058] FIG. **6B** illustrates an example HUD image **70A** corresponding to the detected movement of FIG. **6A** that can be displayed to provide the notification discussed above. Referring to FIG. **6B**, an avatar **58A** is provided of the commercial vehicle **10**, and a notification **60A**, which is an arrow, is provided to indicate what direction the detected object **52** is moving with respect to the commercial vehicle **10**. In one example, as the detected object **52** moves right, only a rightward arrow is provided, and as the detected object **52** rounds the front right corner of the commercial vehicle **10** and starts moving towards the trailer **14** of the commercial vehicle **10**, a rearward arrow is provided (e.g., the curved rearward arrow **60A**). A HUD display area **62** may be allocated to displaying other information, such as a vehicle speed.

[0059] FIG. **6C** illustrates another example HUD image **70B** corresponding to the detected movement of FIG. **6A**. Referring to FIG. **6C**, the avatar **58A** is provided of the commercial vehicle **10**, and a notification **60B** is provided to indicate what direction the detected object **52** is moving with respect to the commercial vehicle **10**. A plurality of discrete display elements **72** (e.g., groups of pixels) are arranged into a rectangular shape around the commercial vehicle. The notification **60B** includes activating a series of the display elements in a sequence (e.g., sequentially illuminating the order of boxes labeled A, B, C, D, E) to indicate the direction of movement of the detected object **52**. Here too, rightward movement of the object **52** could be detected and indicated by sequentially illuminating elements A, B, C, and once the detected object **52** rounded the front right corner of the commercial vehicle **10**, rearward directionality could be indicated (e.g., by

illuminating display elements CDE. Another example is illuminating the entire sequence A, B, C, D, E).

[0060] FIG. 7A provides a schematic birds-eye view of the commercial vehicle **10**, and illustrates how a detected object **52** moves with respect to the commercial vehicle **10** from a first position **54A** in front of the commercial vehicle **10** (in field of view FOV.sub.EX2) to a second position **54B** on a right side of the commercial vehicle (in the field of view FOV.sub.EX4).

[0061] FIG. 7B illustrates an example HUD image **70C** corresponding to the detected movement of FIG. 6A. In the example of FIG. 7B, an avatar **58B** is provided that depicts a pedestrian VRU, and a notification **60C**, which is a curved arrow, is provided to indicate what direction the detected object **52** is moving with respect to the commercial vehicle **10**. As shown, the arrow is a curved arrow that points downward and to the left, to indicate that the VRU is rounding a right rear corner of the commercial vehicle **10**. Here too, rearward directionality could be detected (and correspondingly only a downward pointing arrow indicating rearward movement with respect to the commercial vehicle **10**) may be indicated, and once the detected object **52** rounded the right rear corner of the commercial vehicle **10**, leftward directionality could be indicated (e.g., with the curved arrow shown in FIG. 7B).

[0062] FIG. 7C illustrates another example heads up display (HUD) image corresponding to the detected movement of FIG. 6A.

[0063] As discussed above, the CMS display **18C** may be used for the rear camera **20D**. As the VRU enters the field of view FOV.sub.EX4 of the commercial vehicle **10**, a notification **60D** may be provided that indicates which of the plurality of CMS displays **18** the object is predicted to appear on next. Thus, if the VRU **52** is heading rearward, and optionally also if the VRU **52** starts to round the rear right corner of the commercial vehicle **10** as shown in FIG. 7A), notification **70C** may be provided that tells a vehicle occupant to look upwards away from display **18B** towards CMS display **18C** so that the VRU can be viewed on CMS display **18C**.

[0064] Thus, the notifications provided by the CMS ECU **22** may include shapes that indicate direction (e.g., arrows), and/or or display elements activated in a sequence to indicate a direction.

[0065] In one or more embodiments, the notification includes an indication of how close the moving object is to the commercial vehicle **10** and/or an indication of how likely a collision is between the commercial vehicle **10** and the moving object.

[0066] FIG. 8 schematically illustrates a scenario where the commercial vehicle **10** is at a work site, such as a warehouse or a mining site, and is backing up (see arrow **75**). An object **74** is detected, which may be a forklift, that is moving with respect to the commercial vehicle **10** and with respect to an environment surrounding the commercial vehicle **10**. In the case of FIG. 8, the detected object **74** and is moving towards a backup path of the commercial vehicle **10** (see arrow **76**). In such an example, the ECU **22** may provide on the HUD **18E** an indication of how close the moving object **74** is to the commercial vehicle **10** (i.e., an estimated distance between the commercial vehicle **10** and the detected object **74**) and/or may provide an indication of how likely a collision is between the commercial vehicle **10** and the moving object **74** is at the current trajectories of the commercial vehicle **10** and moving object **74**.

[0067] FIG. 9 is a flowchart of an example method **100** for a CMS **15**. The method may be performed by the CMS ECU **22**, for example. A plurality of CMS cameras **20** are used to record images of an environment surrounding a commercial vehicle **10** (step **102**). Image feeds from the plurality of CMS cameras **20** are displayed a plurality of CMS displays **18** (step **104**). An object outside of the commercial vehicle **10** is detected that is moving with respect to the commercial vehicle **10** and with respect to the environment surrounding the commercial vehicle **10** (step **106**). A notification of the object is displayed on a HUD **18E** (step **108**). The notification indicates at least one of direction the detected object is moving with respect to the commercial vehicle **10**, and which of the plurality of CMS displays **18** the object is predicted to appear on next.

[0068] As discussed above, to perform step **106**, the CMS ECU **22** may utilize conventional object

detection techniques (e.g., image processing and/or processing of data from the object detection sensors **30**) to detect that an object is moving with respect to the commercial vehicle **10** and an environment surrounding the commercial vehicle **10**.

[0069] In one or more embodiments, the CMS ECU **22** is configured to determine that the moving object is outside of, but moving towards, a field of view of a particular one of the plurality of CMS cameras **20**. The notification provided may indicate the particular one of the plurality of CMS displays **18** the object is predicted to appear on next (e.g., CMS display **20C**), which may be the CMS display (e.g., display **20D**) that provides the image feed from the particular one of the CMS cameras (as discussed above in connection with FIG. **7A**).

[0070] In one example, in step **108**, different notifications may be provided for different types of detected objects (e.g., a first notification for a VRU, a second notification that is different than the first notification for a non-VRU). In one or more embodiments,

[0071] In one example, the commercial vehicle **10** is a mining vehicle or a construction vehicle, and the detected object is a non-mining vehicle (e.g., a truck) or a VRU.

[0072] In one example, the commercial vehicle is a construction vehicle (e.g., a cement mixer) and the detected object is a non-construction vehicle (e.g., a truck) or VRU. or

[0073] The various features discussed herein provide a variety of ways of integrating a HUD **18E** with a CMS **15**. The features provide useful notifications to a vehicle occupant by directing them to regions of interest and in some embodiments also providing collision warnings, while also minimizing distraction of the vehicle occupant.

[0074] Although example embodiments have been disclosed, a worker of ordinary skill in this art would recognize that certain modifications would come within the scope of the claims. For that reason, the following claims should be studied to determine their true scope and content.

Claims

1. A camera monitor system (CMS), comprising: a plurality of CMS cameras configured to record images of an environment surrounding a commercial vehicle; a plurality of CMS displays; a heads-up display (HUD); and processing circuitry configured to: display image feeds from the plurality of CMS cameras on the plurality of CMS displays; detect an object outside of the commercial vehicle that is moving with respect to the commercial vehicle and with respect to the environment surrounding the commercial vehicle; and display a notification of the object on the HUD, the notification indicating at least one of a direction the object is moving with respect to the commercial vehicle, and a particular one of the plurality of CMS displays the object is predicted to appear on next.
2. The CMS of claim 1, wherein the processing circuitry is configured to detect the object based on one or more of the image feeds.
3. The CMS of claim 1, wherein the CMS includes: at least one object detection sensor comprising at least one of a radar sensor, a lidar sensor, and an ultrasonic sensor; wherein the processing circuitry is configured to detect the object based on output from the at least one object detection sensor.
4. The CMS of claim 1, wherein the processing circuitry is configured to display a first notification on the HUD based on the object being a VRU, and display a second notification on the HUD based on the object being a non-VRU, the second notification being different than the first notification.
5. The CMS of claim 1, wherein the object is a vulnerable road user (VRU).
6. The CMS of claim 1, wherein the object is a forklift.
7. The CMS of claim 1, wherein: the commercial vehicle is a mining vehicle or a construction vehicle; and the object is a vehicle that is not a mining vehicle or a construction vehicle.
8. The CMS of claim 1, wherein the processing circuitry is configured to: determine that the object is outside of, but is moving towards, a field of view of a particular one of the plurality of CMS

cameras; wherein the notification indicates said particular one of the plurality of CMS displays the object is predicted to appear on next; and wherein said particular one of the plurality of CMS displays is configured to display the image feed from said particular one of the CMS cameras.

9. The CMS of claim 1, wherein the notification includes: an indication of how close the detected object is to the commercial vehicle; or an indication of how likely a collision is between the commercial vehicle and the detected object.

10. The CMS of claim 1, wherein the notification comprises: a plurality of display elements that are activated in a sequence to indicate the direction; or a shape that indicates the direction.

11. A method for a camera monitor system (CMS), comprising: utilizing a plurality of CMS cameras to record images of an environment surrounding a commercial vehicle; displaying image feeds from the plurality of CMS cameras on a plurality of CMS displays; detecting an object outside of the commercial vehicle that is moving with respect to the commercial vehicle and with respect to the environment surrounding the commercial vehicle; and displaying a notification of the object on a heads up display (HUD), the notification indicating at least one of a direction the object is moving with respect to the commercial vehicle, and which of the plurality of CMS displays the object is predicted to appear on next.

12. The method of claim 11, said detecting comprises detecting the object based on one or more of the image feeds.

13. The method of claim 11, wherein said detecting comprises detecting the object based on output from at least one object detection sensor, the object detection sensor comprising at least one of a radar sensor, a lidar sensor, and an ultrasonic sensor.

14. The method of claim 11, wherein said displaying includes: displaying a first notification on the HUD based on the object being a VRU; and displaying a second notification on the HUD based on the object being a non-VRU, the second notification being different than the first notification.

15. The method of claim 11, wherein the object is a vulnerable road user (VRU).

16. The method of claim 11, wherein the object is a forklift.

17. The method of claim 11, wherein: the commercial vehicle is a mining vehicle or a construction vehicle; and the object is a vehicle that is not a mining vehicle or a construction vehicle.

18. The method of claim 11, wherein method includes: determining that the object is outside of, but is moving towards, a field of view of a particular one of the plurality of CMS cameras; wherein the notification indicates said particular one of the plurality of CMS displays the object is predicted to appear on next; and wherein said particular one of the plurality of CMS displays is configured to display the image feed from said particular one of the CMS cameras.

19. The method of claim 11, wherein the notification includes: an indication of how close the detected object is to the commercial vehicle; or an indication of how likely a collision is between the commercial vehicle and the detected object.

20. The method of claim 11, wherein the indication of what direction the object is moving comprises: a plurality of display elements that are activated in a sequence to indicate the direction; or a shape that indicates the direction.
